

What is a Prototype in JavaScript?

(Easy Explanation)

Quick Example



```
1  const user = {
2    sayHello() {
3      console.log("Hello!");
4    },
5  };
6
7  // admin uses user as its prototype
8  const admin = Object.create(user);
9
10 admin.sayHello(); // Output: Hello!
11
```

Even though `admin` doesn't have a `sayHello()` method, it gets it from `user` — because `user` is its prototype.

Custom Method Example:

Add `addition()` to Arrays

Let's say arrays in JavaScript don't have a method to add all numbers.

You can create your own method using prototype:



```
1  Array.prototype.addition = function () {  
2    return this.reduce((sum, value) => sum + value, 0);  
3  };  
4  
5  const nums = [5, 10, 15];  
6  console.log(nums.addition()); // Output: 30  
7
```

What are Prototypes in JavaScript?

1. **In JavaScript, every object has a hidden link to another object — this is called its prototype.**
2. **You can think of the prototype as a backup helper.**
3. **If JavaScript doesn't find a method or property in the object itself, it looks in its prototype — and keeps going up the chain until it finds it (or gives up).**
4. **This idea is called the "prototype chain", and it helps objects share code.**



Real-Life Example (Backup Storage)

1. **Think of a vending machine:**

2. You press the "Chips" button.

3. **That specific machine doesn't have chips →**



4. It checks its backup storage (its prototype).

5. **If chips are there →**  **you get your snack.**

6. If not, it checks another backup... and so on.

7. That's exactly how JavaScript prototype chain works!



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